

Agent Patterns Quick Reference

Five essential patterns for building effective AI agents with Claude

RALPH Loop Pattern

React, Act, Learn, Plan, Handle — a cyclical agent workflow.

Aspect	Description
When to Use	Long-running tasks requiring continuous adaptation
Key Steps	React to feedback, act on decisions, learn from outcomes, plan next iteration, handle errors
Pitfalls	Infinite loops, poor learning signals, unclear decision criteria

Beads Pattern

Decompose complex tasks into independent, sequential subtasks.

Aspect	Description
When to Use	Breaking large projects into manageable, discrete steps
Design Beads	Clear inputs/outputs, single responsibility, testable independently
Commit Strategy	One commit per completed bead for clear history

Multi-Phase Planning

Structured approach with planning, analysis, design, implementation, and validation.

Phase	Focus
1. Plan	Define objectives, constraints, success criteria
2. Analyze	Examine current state, identify risks, gather requirements
3. Design	Create architecture, define interfaces, plan implementation
4. Implement	Execute plan, build components, integrate systems
5. Validate	Test, verify quality, confirm success criteria met

Checkpoint Strategy: Review at each phase boundary before proceeding.

Subagents Pattern

Delegate specialized tasks to focused agents with specific capabilities.

Pattern	Usage
Sequential	Pass output from one subagent as input to next
Parallel	Run multiple subagents simultaneously, merge results
Hierarchical	Parent agent orchestrates child agents at different levels

Delegation Tips: Be explicit about subagent scope, expectations, and success criteria.

Agent Swarms Pattern

Multiple agents work collaboratively, communicating through shared files and context.

When to Use:

- Large-scale projects requiring diverse expertise
- Complex problems needing multiple perspectives
- Systems requiring continuous optimization

Communication via Shared Files:

- Use shared config files for coordination
- Maintain shared log/state files for visibility
- Implement message queue pattern for async communication

Merge Strategy:

- Designate lead agent for final integration
- Use version control for conflict resolution
- Define clear merge criteria and validation rules